

AY: 2025-2026

Course: PPOM 1

Class of 2029

Week 14, Lecture 135

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DPR Lecture:

History and Physical exam of the Ankle and Foot

Presented by:

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Session Objectives

- Demonstrate the components of the history taking and physical examination of the ankle and foot
- Describe subjective and objective findings relating to the examination of the ankle and foot
- Distinguish Normal vs. Abnormal examination findings
- Recall the appropriate way to document common normal and abnormal findings of the history and physical Examination
- Incorporation of OMM in a Case Study

RED WOLVES and BEARS

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The Ankle and Foot

- The total weight of the body is transmitted through the ankle to the foot
- Must balance the body and absorb the impact of the heel strike and gait
- Frequent sites of sprain and bony injury

The Ankle and Foot

- Principal Joints of the Ankle
 - Tibiotalar joint
 - Subtalar (Talocalcaneal) joint
- Principal Landmarks of the Ankle
 - Medial Malleolus
 - Lateral Malleolus
 - Calcaneus

The Ankle and Foot

- Longitudinal arch
 - Spans the foot, extending from the calcaneus along the tarsal bones, to the metatarsals and toes
 - Subtalar (Talocalcaneal) joint
- Hindfoot
 - Calcaneus and talus
 - Contains the subtalar joint formed by the articulation of these bones
- Midfoot
 - Navicular, cuboid, and 3 cuneiform bones
 - Calcaneocuboid and talocalcaneonavicular joint allow for inversion and eversion of the foot

The Ankle and Foot

- Forefoot
 - 5 metatarsal bones and all of the phalanges
 - Tarsometatarsal joints
 - 5 Metatarsals articulate with the 5 proximal phalanges forming the metatarsophalangeal joints
 - Toes 2→5 The articulations between the proximal and middle phalanges (PIP joint)s and the middle and distal phalanges (DIP joint)
 - First Toe
 - Interphalangeal joint and metatarsophalangeal joint

The Ankle and Foot

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Medial View



Lateral View

The Ankle and Foot

- Muscles of the foot and ankle
 - Extrinsic
 - Anterior Compartment of the lower leg
 - Ankle Dorsiflexors and toe extensors
 - Tibialis anterior → Dorsiflexion of ankle
 - Extensor digitorum longus → Extension of toes 2-5
 - Extensor hallucis longus → Extension of first toe
 - Intrinsic

Sources: Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16;
Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports
Medicine; Chapter 9

The Ankle and Foot

- Muscles of the foot and ankle
 - Extrinsic
 - Posterior (superficial and deep) compartment of the lower leg
 - Ankle plantarflexor
 - Gastrocnemius, Soleus, and Plantaris (superficial posterior compartment)
 - Toe Flexors (deep posterior compartment)
 - Flexor hallucis longus inserts at the first distal phalanx
 - Medial to the FHL is the Flexor digitorum longus (FDL) inserts at the distal phalanges of toes two to five
 - Tibialis posterior (most medial of the muscles)
 - Inserts on the navicular, cuneiform and cuboid bones and the 3 middle metatarsal bases
 - Intrinsic

Sources: Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16;
Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports
Medicine; Chapter 9

The Ankle and Foot

- Muscles of the foot and ankle
 - Extrinsic
 - Lateral compartment of the lower leg
 - Peroneus longus and brevis muscles
 - Foot eversion and some ankle plantarflexion
 - Peroneus longus
 - Inserts at the medial cuneiform and first metatarsal
 - Peroneus brevis
 - Inserts on the base of the fifth metatarsal
 - Intrinsic
 - Distal to the ankle within the plantar surface of the foot
 - Additional movements of the toes
 - Maintain the arch of the foot

The Ankle and Foot – The Medical History

- Acute vs. Chronic
- Ask about the nature and circumstance of the injury
- Position of the foot and ankle at the time of injury
- Was there a “Snap” or “Pop” heard or felt
- Immediate pain and swelling?
- “Point to the pain”
- OPQRST

The Ankle and Foot – The Medical History

- Is this a chronic injury that was once acute and not healed?
- How much activity is the patient performing per day? Per Week? How much time or how many miles
- Ask about footwear used, and long have they been using their current shoes
- Progression of pain

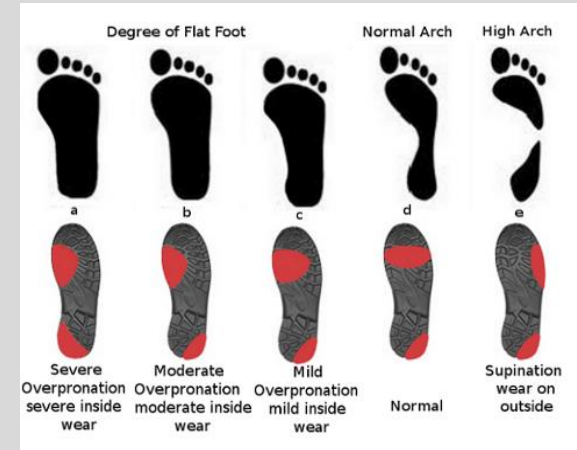
The Ankle and Foot – The Physical Exam

- Inspection
 - Observe ambulation
 - Shoes and socks should be removed
 - Note any swelling, erythema, or ecchymosis around the foot or ankle
 - Bony landmarks should be visible
 - Inspect skin for lesions
 - Blisters, ingrown toe nails, and distal toe discoloration may indicate poor fitting in shoes
 - Look at the shoes and the wear patterns

The Ankle and Foot – The Physical Exam

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The Ankle and Foot – The Physical Exam

- Palpation
 - Begin just below the knee and palpate along the entire length of the lower leg
 - Palpate the proximal fibula and tibia and follow the shafts of these bones down to the respective malleoli
 - Palpate bones and ligaments looking for tenderness or instability

The Ankle and Foot – Palpation

- Bilateral on skin exam
- Use fingerpads or thumbs for palpation



- Landmarks
 - Lateral/Medial malleolus
 - Calcaneus, Talar head
 - Navicular, Cuboid
 - First metatarsal bone
 - First metatarsal phalangeal joint
 - Styloid process of 5th metatarsal
 - Deltoid ligament
 - Anterior talofibular ligament
 - Calcaneofibular ligament
 - Achilles tendon

The Ankle and Foot – Range of Motion

- Range of Motion (ROM)
 - Active
 - Passive
 - Should be equal bilaterally
 - Recorded in degrees

The Ankle and Foot – Range of Motion

Ankle and Foot Movement	Primary Muscles Affecting Movement	Patient Instructions
Ankle Flexion (Plantar Flexion)	Gastrocnemius, soleus, plantaris, tibialis posterior	<i>“Point your foot toward the floor.”</i>
Ankle Extension (Dorsiflexion)	Tibialis anterior, extensor digitorum longus, and extensor hallucis longus	<i>“Point your foot toward the ceiling.”</i>
Inversion	Tibialis posterior and anterior	<i>“Bend your heel inward.”</i>
Eversion	Peroneus longus and brevis	<i>“Bend your heel outward.”</i>

The Ankle and Foot – Range of Motion

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Forefoot

- Bilateral exam
- On skin exam
- Passive Range of Motion



Inversion



Eversion

Ankle



Inversion



Eversion

The Ankle and Foot – Range of Motion

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- Bilateral exam
- On Skin
- Active Range of Motion
 - Dorsiflexion
 - Plantarflexion
 - Eversion
 - Inversion



Plantarflexion



Dorsiflexion



Eversion



Inversion

The Ankle and Foot – Strength Testing

- Bilateral exam
- On Skin
- Active Strength Testing against resistance applied by the examiner

Plantarflexion



Dorsiflexion



Anterior Drawer Test

- Examination
 - Affected foot off table
 - Examiner cups heel and places other hand on the anterior tibia
 - A posterior to anterior force is applied
 - Test for Anterior Talofibular Ligament integrity
- Positive Test
 - Ankle Translates anteriorly without firm endpoint
- Indicates
 - Anterior Talofibular Ligament tear / disruption with laxity
 - Mild Sprain with pain alone

Sources:

1. ICC Guide
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 9

Lateral Ankle Stability Test (Talar Tilt Test)

- Examination
 - Same position as in the Anterior Drawer Test
 - Stabilizing hand of the examiner remains in the same position
 - Examiner grasps either the dorsum of the foot or the inferior calcaneus
 - Talus is inverted to examine range of motion
 - Test for ATFL and CFL integrity
- Positive Test
 - Presence of laxity / increased range of motion (excessive tilt) or pain as compared to the unaffected side.
- Indicates
 - Anterior Talofibular and Calcaneofibular Ligament tear / disruption with laxity
 - Mild Sprain with pain alone

Sources:

1. ICC Guide
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 9

Eversion Test

- Examination
 - Same position as in the Anterior Drawer Test
 - Stabilizing hand of the examiner remains in the same position
 - Examiner grasps either the dorsum of the foot or the inferior calcaneus
 - The foot is everted to examine range of motion
 - Test for Deltoid ligament
- Positive Test
 - Presence of laxity / increased range of motion or pain at end ankle ROM, as compared to the unaffected side.
- Indicates
 - Deltoid ligament tear / disruption
 - Mild Sprain with pain alone

Source: Stockard, Alan, Osteopathic Clinical Joint Exam;
American Osteopathic Academy of Sports Medicine;
Chapter 9

External Rotation (Kleiger's Test)

- Examination
 - Patient seated
 - Stabilizing hands of the examiner on the anterior tibia
 - Examining hand externally rotates the foot against the stabilized lower leg
 - The talus should be included in the rotation
- Positive Test
 - Production of pain along the syndesmosis or medial ankle
- Indicates
 - Production of pain along the syndesmosis (Tear/disruption of the syndesmosis)
 - Production of pain along the medial ankle (Tear / Disruption of the medial deltoid ligaments)

Source: Stockard, Alan, Osteopathic Clinical Joint Exam;
American Osteopathic Academy of Sports Medicine;
Chapter 9

(Syndesmosis) Squeeze Test

- Examination
 - Examiner places the thenar eminence of one hand against the tibial shaft
 - The thenar eminence of the other hand against the fibular shaft
 - The examiner squeezes for 2-3 seconds followed by a brisk release
- Positive Test
 - Production of pain at the level of the syndesmosis
- Indicates
 - Syndesmosis tear / disruption

Thompson Test

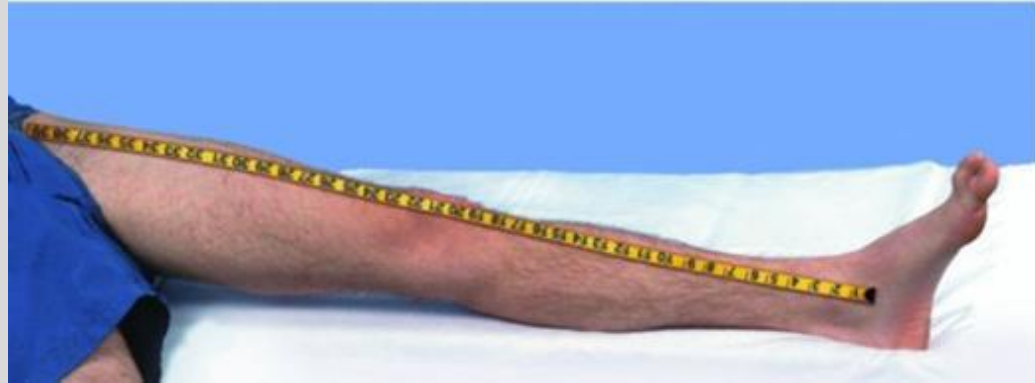
- Examination
 - Patient in a prone position with the distal 1/3 of lower leg hanging off the table
 - The examiner squeezes the gastrocnemius muscle and observes for plantarflexion of the ankle
 - Tests for Achilles tendon rupture
- Positive Test
 - Absence of plantar flexion of the ankle with the maneuver
- Indicates
 - Rupture of the Achilles tendon

Tibiotalar Shuck Test (Cotton's Test)

- Examination
 - Stabilizing hand holding the lower leg
 - Examining hand grasps the talus from underneath, moving it medially and laterally
- Positive Test
 - Production of pain and/or laxity as compared to the non-affected side
- Indicates
 - Syndesmosis injury / disruption (High Ankle Sprain)

The Ankle and Foot – Measuring Leg Length

- Patient should be relaxed in the supine position and symmetrically aligned with legs extended
- With a tape, measure the distance between the ASIS and the Medial malleolus
- The Tape should cross the knee on its medial side

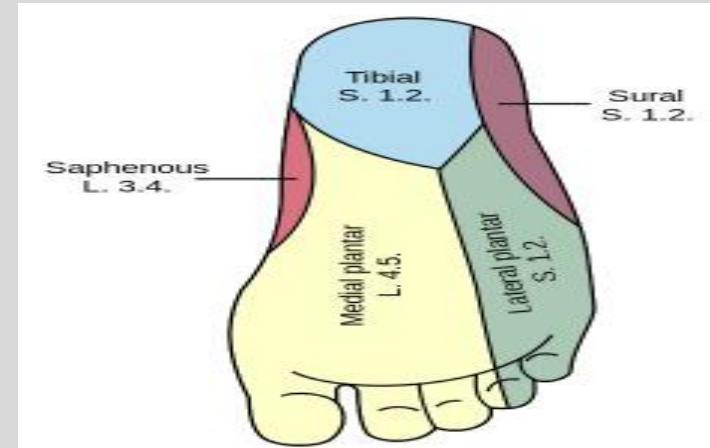


The Ankle and Foot – Osteopathic Exam

- Osteopathic Evaluation
 - Dysfunction
 - Lymphatics
 - Restrictions
 - Popliteal fossa

The Ankle and Foot – Neurovascular Exam

- Neurovascular Examination
 - Deep tendon reflexes
 - L4 patellar, no L5, S1 Achilles
 - Sensory
 - L4 medial foot, L5 dorsal foot, S1 lateral foot
 - Vascular
 - Dorsal pedis
 - Posterior tibial



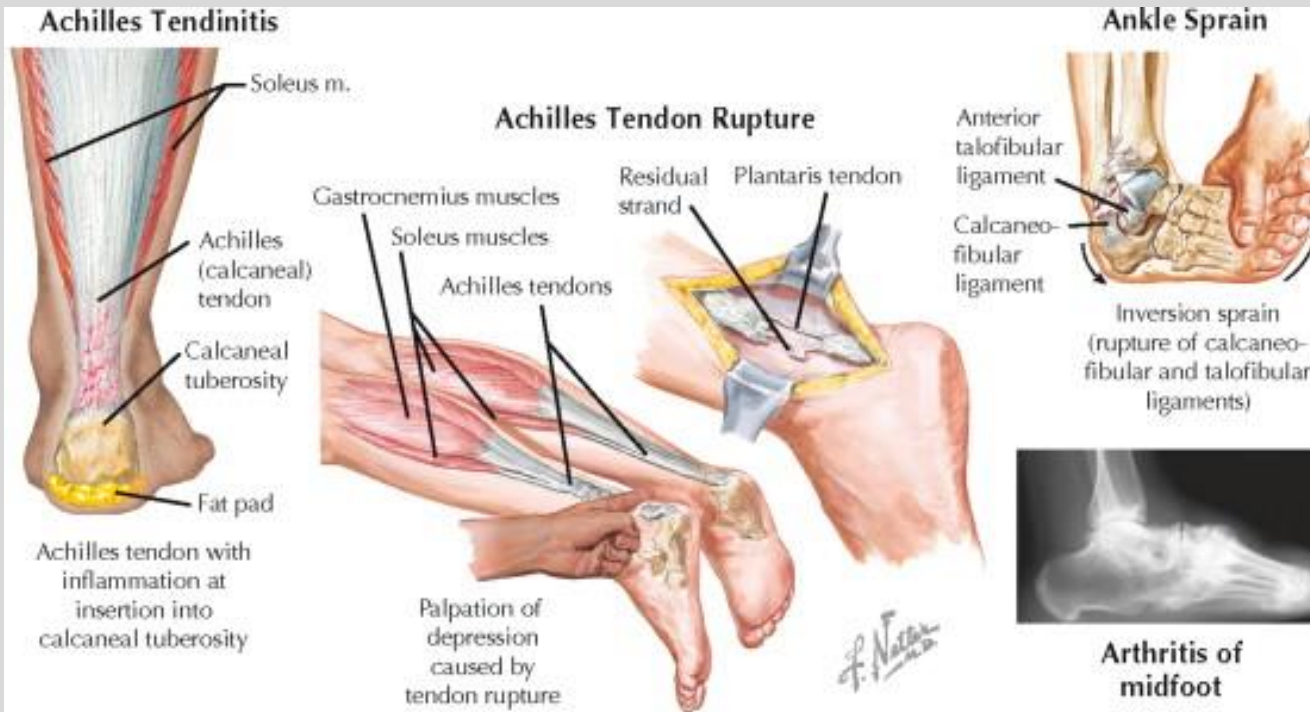
The Ankle and Foot

- Common Conditions
 - Lateral/Medial Ankle Sprains
 - Tendinitis
 - Gastrocnemius/Soleus Strain
 - Plantar Fasciitis
 - Chronic Exertional Compartment Syndrome
 - Medial Tibial Stress Syndrome

The Ankle and Foot

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Achilles Tendinitis

- Occurs at or above insertion of Achilles tendon
- Micro-trauma to insertion
- Exam shows heel pain, worse with push off; tender to palpation
- Treatment includes rest, Ice, NSAIDs, exercises, heel lift
- Avoid Cortisone shots

Case Study

- 22 y/o male presents c/o right ankle injury. He states that he was playing basketball several hours ago, jumped up and landed on his right ankle which "twisted". Now he c/o "very painful and swollen ankle" and states that he is unable to bear weight due to the pain. He went to his old trainers office on campus and was given ice to put on ankle and advised to come here for further evaluation.

Case Study

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- Patient is a 2nd year medical student
- He was scheduled to be at genetic lecture

Case Study

Physical Exam:

Vital: BP 110/90; HR 64; RR 14; T 99.1; Weight:
Unable to obtain due to non-weightbearing status

General: Pleasant male, in mild distress secondary to
pain. AA&O x 3

HEENT: NC/AT

Case Study

Physical Exam (continued):

Extremity: The right ankle was examined and compared to the left ankle. Unable to bear weight secondary to pain. There is no obvious bone deformity or dislocation. There is no muscle malformation. There is significant swelling and ecchymosis mostly along the lateral malleolus. There is no erythema. There is tender to palpation at the lateral malleolus. No other bony tenderness. There is no step-off. Range of motion is decreased in all direction secondary to pain. Unable to perform provocative tests and maneuvers secondary to pain.

Case Study

- **Assessment:**
 - Right Ankle Sprain
 - Fracture right distal fibula
 - High Ankle Sprain
 - Achilles Tendinitis
 - Ankle Dislocation

Case Study

- Plan:
 - 1. X-ray right ankle ordered (3 views)
 - 2. Right ankle wrapped in office with 4-inch coban wrap
 - 3. Ankle placed in Aircast for stability
 - 4. Crutches dispense in office, patient fitted and counseled on proper use
 - 5. Advised patient to stay off ankle until x-rays are taken.

Case Study

- Plan:
 - 6. Discuss application of RICE treatment
 - 7. OTC NSAIDs as needed for pain
 - 8. Plan to follow up in office tomorrow morning to review x-rays

Ankle X-rays

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Source: Real Case at NYIT

Ankle Sprains

- Greatest risk factor for ankle sprain is a previous ankle sprain
- Most ankle sprains are injuries to the lateral ligaments of the ankle due to inversion of the ankle with some degree of plantar flexion
- Tenderness, swelling, and ecchymosis common

Ankle Sprains

- Grading of lateral ankle sprains
- Tests of ligament integrity are difficult to perform at time of acute injury

Ottawa Ankle and Foot Rules

- Well-established clinical guidelines used to determine the need for radiography
- Ankle Radiography is needed
 - Pain in the malleolar zone
 - Plus either
 - bone tenderness over areas of potential fracture
 - Inability to bear weight for four steps immediately after the injury and in the emergency department or physician's office

Ottawa Ankle and Foot Rules

- Foot Radiography is needed
 - Pain in the midfoot zone
 - Plus either
 - bone tenderness over areas of potential fracture
 - Base of the fifth metatarsal or the navicular bone of the midfoot
 - Inability to bear weight for four steps immediately after the injury and in the emergency department or physician's office

High (Syndesmotic) Ankle Sprain

- Tibiofibular syndesmosis
 - Anterior tibiofibular ligament
 - Posterior superficial and deep tibiofibular ligaments
 - Interosseous membrane

High (Syndesmotic) Ankle Sprain

- Mechanisms similar to common lateral sprains, although a rotational component to the injury seems to be a common factor
- Diagnosis made by attempting to mechanically separate the distal tibia and fibula (stressing the syndesmosis and eliciting pain proximal to the ankle joint)
- Which test did we discuss to examine for High Ankle Sprain??

High (Syndesmotic) Ankle Sprain

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- Which test did we discuss to examine for High Ankle Sprain??
 - Tibiotalar Shuck Test
 - External Rotation Test
- Additional testing
 - Squeeze test
 - Squeeze the lower leg at the midcalf
 - Crossed-leg test
 - Have the patient cross their legs with the injured leg resting at mid-calf on the knee

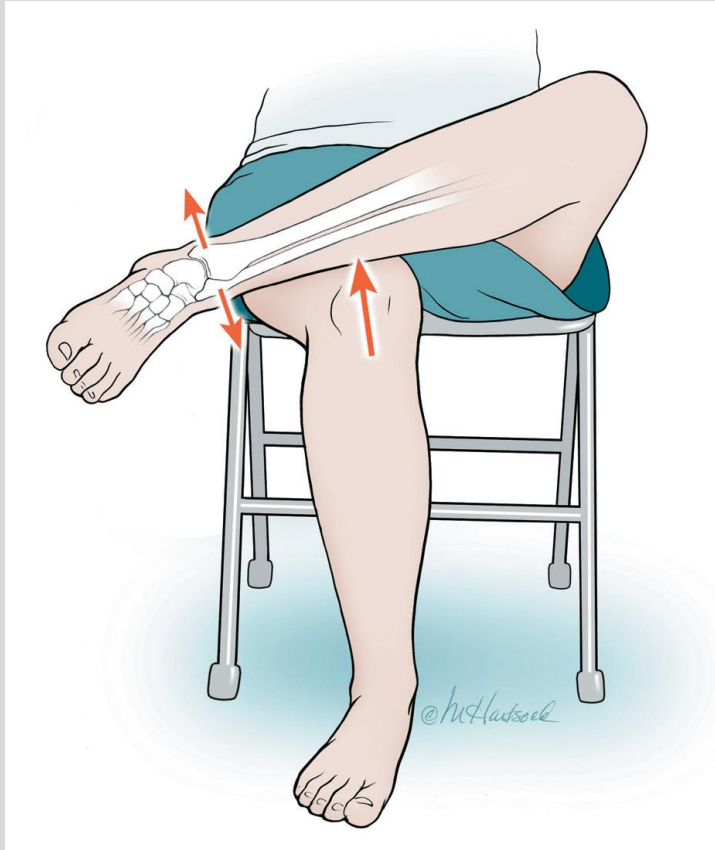


<https://www.nuemblog.com/blog/high-ankle-sprain>

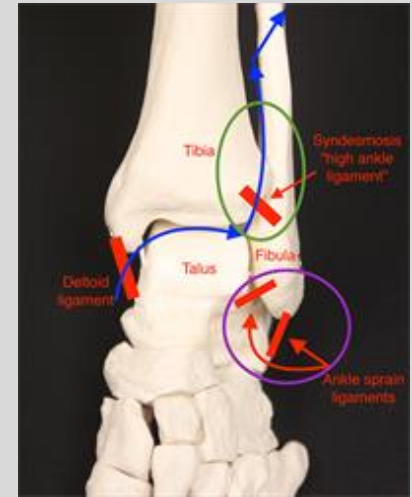
High (Syndesmotic) Ankle Sprain

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<https://www.nuemblog.com/blog/high-ankle-sprain>



<http://www.aofas.org/footcaremd/conditions/ailments-of-the-ankle/Pages/High-Ankle-Sprain.aspx>

Source: Tiemstra, Jeffrey D. Update on Acute Ankle Sprains. *Am Fam Physician*. 2012;85(12):1170-1176.

Ankle Sprains - Treatment

- RICE therapy
- Apply ice on site of injury
- Air stirrup brace or lace-up support
- Early Range of Motion exercises
- Functional Mobilization (bears weight as tolerated)
- Pain medications as tolerated
- OMM

Ankle Sprains - Prevention

- Follow up 4-6 weeks after the injury for assessment of ligamentous instability
- Rehabilitation exercises
 - Proprioceptive training
 - Strengthening
- Supports and Taping

Incorporation of OMM

- Lymphatic congestion of the ankle or foot
- Treatments
 - Improve restrictions
 - Treat the thoracic inlet somatic dysfunction
 - Rib raising
 - Improve restrictions
 - Normalize Sympathetic Activity
 - Improve restrictions at the diaphragms
 - Improve lymphatic flow

Incorporation of OMM

- Lymphatic congestion of the ankle or foot
- Treatments
 - Improve restrictions
 - Improve lymphatic flow
 - Apply the pumps
 - Pedal Pump to improve the lymphatic congestion of the ankle or foot

Incorporation of OMM

- Lymphatic congestion of the ankle or foot
- Treatments
 - Normalizing sympathetic Activity
 - Rib raising
 - Improve restrictions
 - Normalize Sympathetic Activity
 - Soft tissue paraspinal inhibition
 - L1 and L2 sympathetic ganglia is continuous with that of the thoracic paraspinal ganglia.
 - Anterior direct pressure on the erector spinae has the same autonomic effects as rib raising
 - Improve lymphatic flow

Summary Slide

- Demonstrate the components of the history taking and physical examination of the Ankle and Foot
- Describe subjective and objective findings relating to the examination of the ankle and foot
- Distinguish Normal vs. Abnormal examination findings
- Recall the appropriate way to document common normal and abnormal findings of the history and physical Examination
- Incorporation of OMM in a Case Study

Summary Slide

- Core References:

- Bickley, Lynn S.; Szilagyi, Peter G., “Bates' Guide to Physical Examination and History Taking”, 13e, Wolters Kluwer. 2021. Chapter 23: The Musculoskeletal System Subheadings: “The Ankle and Foot”
- Thompson, Jon C. *Netter's Concise Orthopedic Anatomy*, 2nd Edition. 2010 (Chapter 10: “Foot/Ankle”).
- Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 9 “The Foot and Ankle Exam”
- Tiemstra, Jeffrey D. Update on Acute Ankle Sprains. *Am Fam Physician*. 2012;85(12):1170-1176.
- Savarese, Robert. A comprehensive Review in Osteopathic Medicine. 3rd Edition. 2009.

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